Convergence Tests

ANSWER KEY

Section 1: Integral Test and p-Series

1 For what values of p does the p-series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converge?

a. $p > 1$ ✓

b. $p \geq 1$

c. $p < 1$

d. $p > 0$

By the Integral Test, $\int_1^{\infty} x^{-p}\,dx$ converges if and only if $p > 1$. When $p \leq 1$, the p-series diverges.

2 Use the Integral Test to determine the convergence of $\sum_{n=2}^{\infty} \frac{1}{n \ln n}$.

a. **Diverges** ✓

b. Converges to 1

c. Converges to $\ln 2$

d. Inconclusive

$\int_2^{\infty} \frac{1}{x \ln x}\,dx = [\ln(\ln x)]_2^{\infty} = \infty$. Since the integral diverges, the series diverges.

3 Which conditions must $f(x)$ satisfy to apply the Integral Test to $\sum a_n$ where $a_n = f(n)$?

a. f **must be continuous, positive, and decreasing on** $[1, \infty)$ ✓

b. f must be differentiable and alternating

c. f must be bounded and periodic

d. $f'(x) > 0$ everywhere

The Integral Test requires $f(x)$ to be continuous, positive, and eventually decreasing for all $x \geq 1$.

Section 2: Comparison Tests (Direct and Limit)

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4 Determine the convergence of $\sum_{n=1}^{\infty} \frac{1}{n^3+5}$.

a. **Converges by Direct Comparison with $\sum \frac{1}{n^3}$** ✓

b. Diverges by comparison with $1/n$

c. Inconclusive

d. Diverges by the Divergence Test

For $n \geq 1$, $\frac{1}{n^3+5} < \frac{1}{n^3}$. Since $\sum 1/n^3$ is a convergent p-series ($p = 3 > 1$), the series converges.

5 Use the Limit Comparison Test with $b_n = 1/n$ to test $\sum_{n=1}^{\infty} \frac{2n+1}{n^2-3}$.

a. **Diverges because $L = 2 > 0$ and $\sum 1/n$ diverges** ✓

b. Converges because $L = 0$

c. Converges because $L = 2$

d. Inconclusive

$L = \lim_{n\to\infty} \frac{(2n+1)/(n^2-3)}{1/n} = \lim_{n\to\infty} \frac{2n^2+n}{n^2-3} = 2$. Since $0 < L < \infty$, both series share the same behavior. Thus it diverges.

6 Which comparison series b_n is best suited for testing $\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^2+4n}$ via LCT?

a. $b_n = \frac{1}{n^{3/2}}$ ✓

b. $b_n = \frac{1}{n^2}$

c. $b_n = \frac{1}{n}$

d. $b_n = \frac{1}{\sqrt{n}}$

The dominant terms are $\frac{\sqrt{n}}{n^2} = \frac{n^{1/2}}{n^2} = \frac{1}{n^{3/2}}$. Since $p = 3/2 > 1$, it is a convergent p-series.

Section 3: Alternating Series and Error Bounds

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7 What are the two conditions of the Alternating Series Test (AST) for $\sum(-1)^n b_n$ ($b_n > 0$)?

a. $b_{n+1} \le b_n$ **for all** n **and** $\lim_{n\to\infty} b_n = 0$ ✓

b. $\sum b_n$ converges and $b_n > 0$

c. $\lim_{n\to\infty} b_n = 1$ and b_n is decreasing

d. b_n is geometric

The Leibniz criteria require that the terms are positive, monotonically decreasing ($b_{n+1} \le b_n$), and have limit 0.

8 Classify $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$.

a. **Conditionally convergent** ✓

b. Absolutely convergent

c. Divergent

d. Oscillating divergent

The alternating harmonic series converges by AST. However, the series of absolute values $\sum 1/n$ is the divergent harmonic series. Thus it is conditionally convergent.

9 If $S = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^4}$ is approximated by S_3, what is the maximum error by the Alternating Series Estimation Theorem?

a. $|R_3| \le \frac{1}{4^4} = \frac{1}{256}$ ✓

b. $|R_3| \le \frac{1}{3^4} = \frac{1}{81}$

c. $|R_3| \le \frac{1}{5^4}$

d. $|R_3| \le \frac{1}{16}$

By the Alternating Series Estimation Theorem, $|R_n| \le b_{n+1}$. For $n = 3$, $|R_3| \le b_4 = \frac{1}{4^4} = \frac{1}{256}$.

Section 4: Ratio and Root Tests

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10 State the conclusion of the Ratio Test when $L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| < 1$.

a. **The series converges absolutely ✓**

b. The series converges conditionally

c. The series diverges

d. The test is inconclusive

If $L < 1$, the series converges absolutely. If $L > 1$, it diverges. If $L = 1$, the Ratio Test is inconclusive.

11 Apply the Ratio Test to $\sum_{n=1}^{\infty} \frac{3^n}{n!}$.

a. $L = 0 < 1 \Rightarrow$ **Converges absolutely ✓**

b. $L = 3 > 1 \Rightarrow$ Diverges

c. $L = 1 \Rightarrow$ Inconclusive

d. $L = \infty \Rightarrow$ Diverges

$\left|\frac{a_{n+1}}{a_n}\right| = \frac{3^{n+1}}{(n+1)!}\frac{n!}{3^n} = \frac{3}{n+1} \to 0$ as $n \to \infty$. Since $L = 0 < 1$, the series converges absolutely.

12 Use the Root Test to determine the convergence of $\sum_{n=1}^{\infty} \left(\frac{2n+1}{3n-1}\right)^n$.

a. $L = 2/3 < 1 \Rightarrow$ **Converges absolutely ✓**

b. $L = 2/3 \Rightarrow$ Diverges

c. $L = 1 \Rightarrow$ Inconclusive

d. $L = 0$

$\lim_{n\to\infty} \sqrt[n]{|a_n|} = \lim_{n\to\infty} \frac{2n+1}{3n-1} = \frac{2}{3}$. Since $L = 2/3 < 1$, the series converges absolutely.

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